Alkyl Halides

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Reactivity of Organic Halides in Nucleophilic Substitution.

Organic Halides

- R – X is the general formula for organic halides.

  R = alkyl or aryl; X = F, Cl, Br & I
- Most of the organic halides are creatures of the laboratory.
- Only few of them have been isolated from natural products.

Alkyl Halides – Uses

Organohalogen compounds are important for several reason:

- Versatile reagents in organic synthesis.
- Can be converted into alkenes by dehydrohalogenation.
- As solvents (CH_2Cl_2, $CHCl_3$)
- Insecticides, herbicides, fire retardants.
- Refrigerants and Polymers.

Classes

- Primary, Secondary, Tertiary

common organic halides

methylene chloride	CH_2Cl_2
chloroform	$CHCl_3$
carbon tetrachloride	CCl_4
isopropyl chloride	(2-chloropropane)
sec-butyl chloride	(2-chlorobutane)
isobutyl chloride	(1-chloro-2-methylpropane)
tert-butyl chloride	(2-chloro-2-methylpropane)
allyl chloride	(3-chloro-1-propene)
vinyl chloride	(chloroethene)
benzyl chloride	(chloromethylbenzene)
phenyl chloride	(chlorobenzene)

- Dihalegon derivatives (Geminal, Vicinal)
- Preparations

Nucleophilic Substitution

- Nucleophile is an electron rich group.
- Nucleophiles can be classified according to the kind of atom that forms a new covalent bond.

1. Oxygen nucleophile (HO-, CH_3O- …)
2. Nitrogen nucleophiles (NH_3, RNH_2…)
3. Sulfur nucleophiles (HS-, RS- …)
4. Halogen nucleophiles (I- …)

Nucleophilic Substitution

Eg:

- Ethyl bromide reacts with hydroxide ion to give ethanol.
- OH- is the ***nucleophile.***
- Bromide ion is called ***leaving group.***
- One covalent bond is broken and one new bond is formed

$$OH^- + CH_3 - CH_2 - Br \xrightarrow{H_2O} CH_3 - CH_2 - OH + Br^-$$

Nucleophilic Substitution

- If the nucleophile and substrate are ***neutral***, the product will be ***positively*** charged.
- If the nucleophile is a ***negative ion*** and substrate is ***neutral***, the product will be ***neutral***.
- The unshared pair of electron in the nucleophile can be used to make new covalent bond.

$$:Nu + R:L \longrightarrow R:Nu^{+} + :L^{-}$$

$$:Nu^{-} + R:L \longrightarrow R:Nu + :L^{-}$$

Nucleophilic Substitution

- In principle, these reactions may be reversible.
- Because the leaving group also has an unshared pair of electron and that can be used to make a new covalent bond.

$$OH^- + CH_3 - CH_2 - Br \rightleftharpoons CH_3 - CH_2 - OH + Br^-$$

We use various method to force the reaction to go in forward direction.

1. Stronger nucleophile compared to the leaving group.
2. Large excess of one of the reagent
3. Temperature

Nucleophilic Substitution

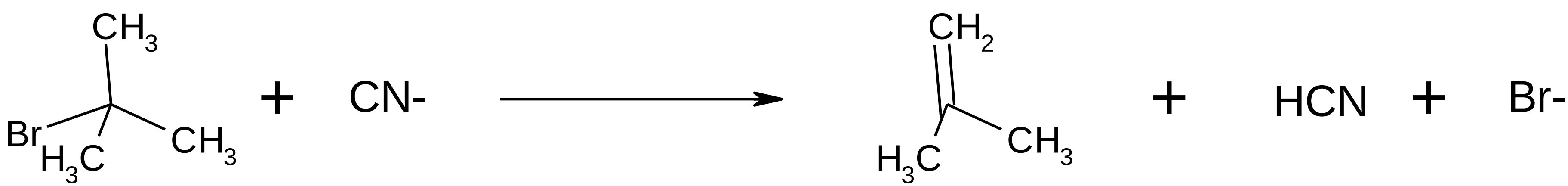
No Nucleophilic Substitution will occur on the carbon which has double bond and triple bond in which leaving group is attached

$$CN^- + CH_3 - CH_2 - CH_2 - Br \longrightarrow CH_3 - CH_2 - CH_2 - CN + Br^-$$

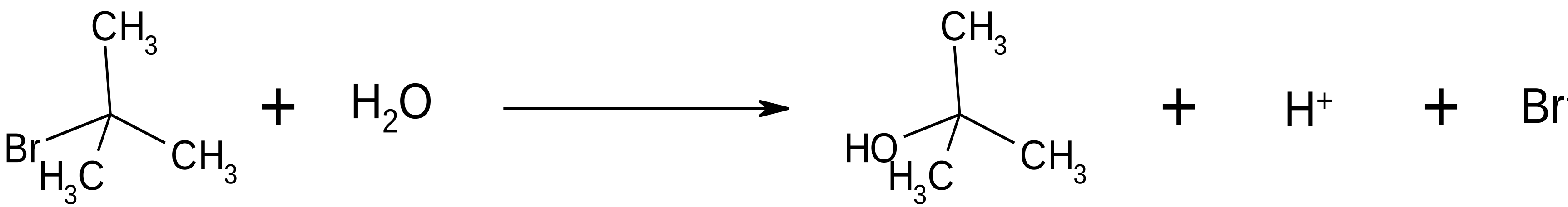
$$H_2C{=}CH\text{-}Br + CN^- \longrightarrow \text{No nuclephilic substitution reaction}$$

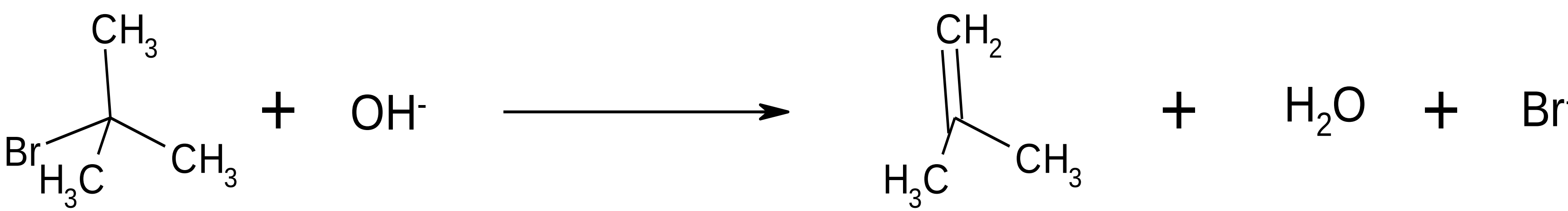
Nucleophile is same but there is difference in carbon which has leaving group.

$$CN^- + CH_3 - CH_2 - CH_2 - Br \longrightarrow CH_3 - CH_2 - CH_2 - CN + Br^-$$



Nucleophile is different but the carbon which has leaving group is same.





S_N Reaction Mechanisms

- "S & N" stands for Nucleophilic Substitution.
- **S_N1 Mechanism**

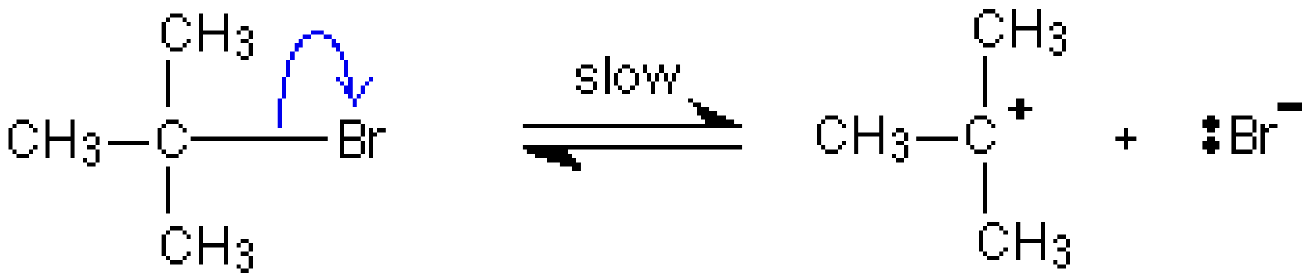
  1 indicate that only one of the two reactant is involved in the rate determining step i.e. formation of carbocation.
- **S_N2 Mechanism**

  2 indicates the reaction is bimolecular i.e. the nucleophile and the substrate are involved in the key step

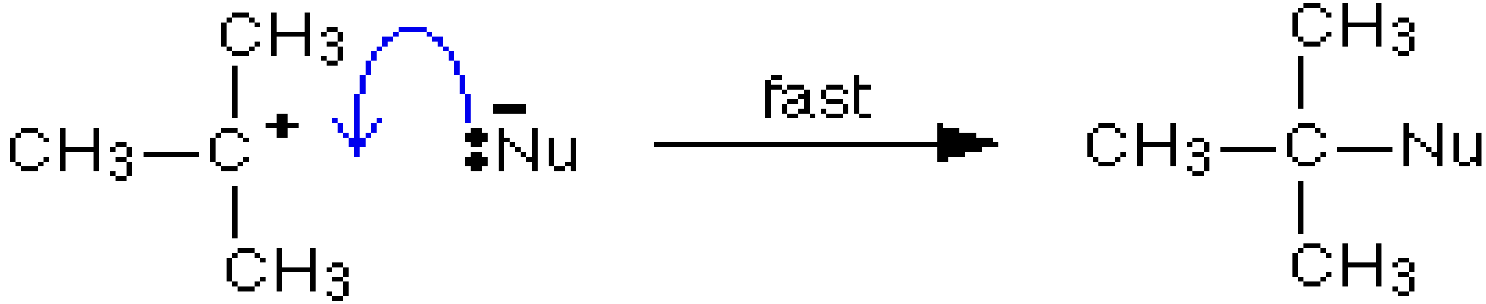
- Primary halides = S_N2
- Secondary halides = both mechanisms!
- Tertiary halides = S_N1
- Depend on: structure of a nucleophile and alkyl halide; the solvent; the reaction temperature…

S_N1 Mechanism

- It's a two step mechanism.
- Formation of the carbocation is first step and which is slow. (Ionization step)



- Second step is the fast i.e. the combination of carbocation with the nucleophile forms product.



- Only one molecular species RX is involve in the heterolytic fission step which is the rate determining step. The reaction is unimolecular (SN1) and the rate R

$$R = K\,(R - X)$$

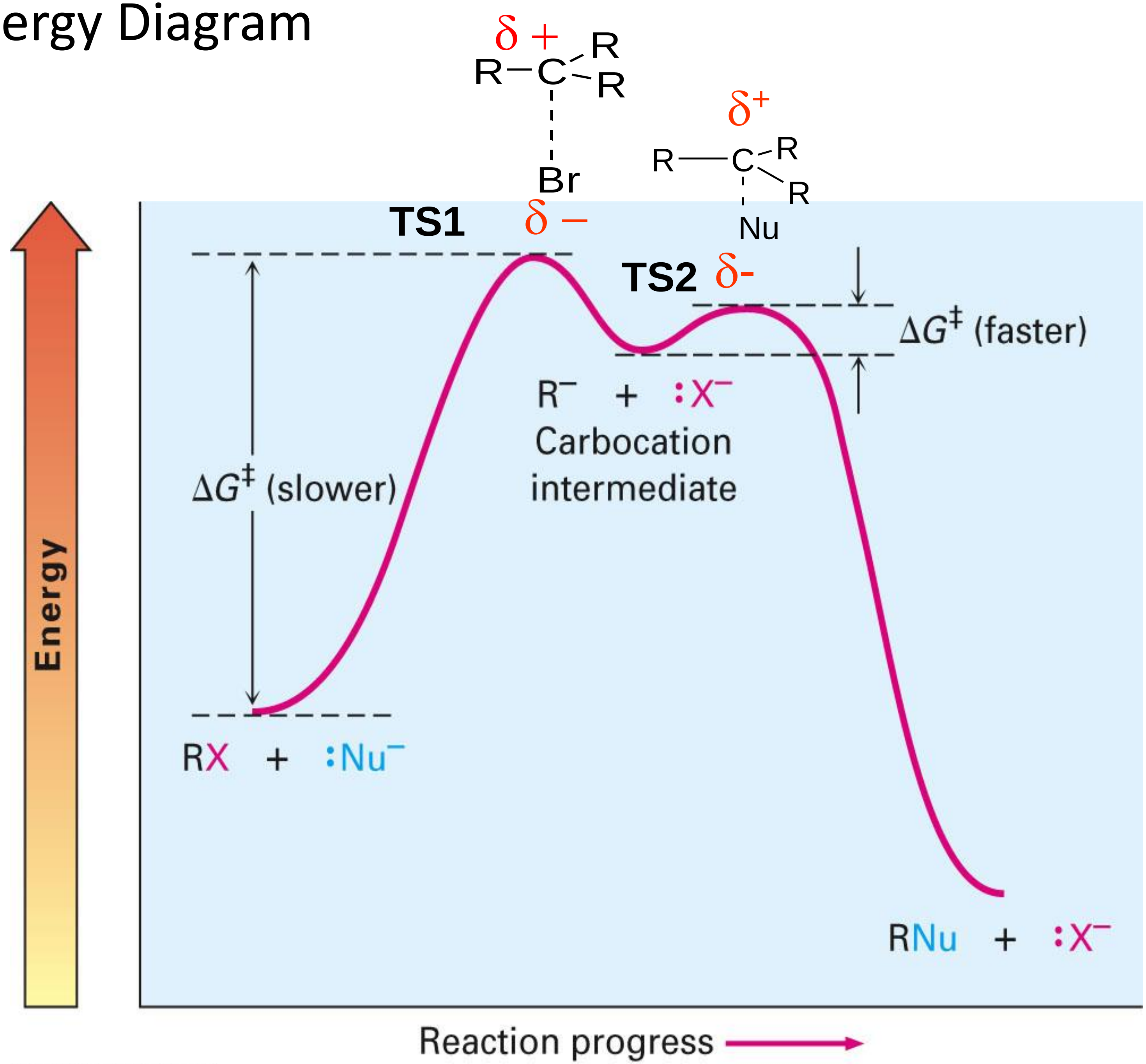
- Number of transition states involved in the reaction

$$TS1 = [\,(CH_3)_3\,C^{\delta +} \ldots\ldots Br^{\delta -}\,]^{+}$$

$$In = (CH_3)_3\,C^{+}$$

$$TS2 = [\,(CH_3)_3\,C^{\delta +} \ldots\ldots Nu^{\delta -}\,]^{+}$$

S_N1 Energy Diagram



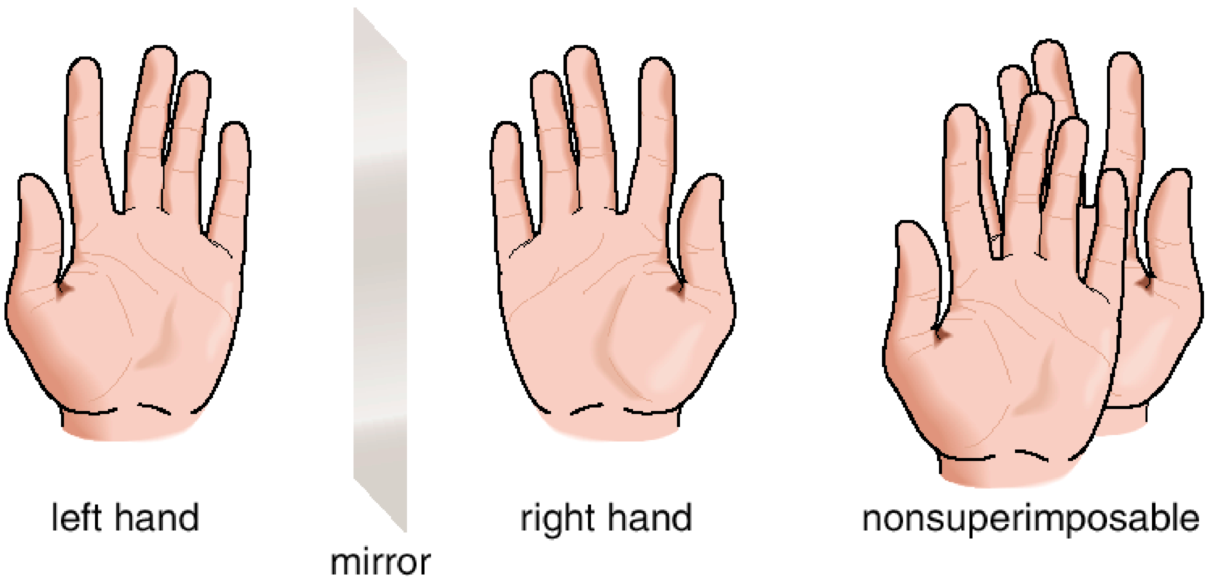
Stereochemistry of SN1 Reactions

Stereochemistry:
Three-dimensional arrangement of atoms (groups) in space

Stereoisomers:
Molecules with the same connectivity but different arrangement of atoms (groups) in space

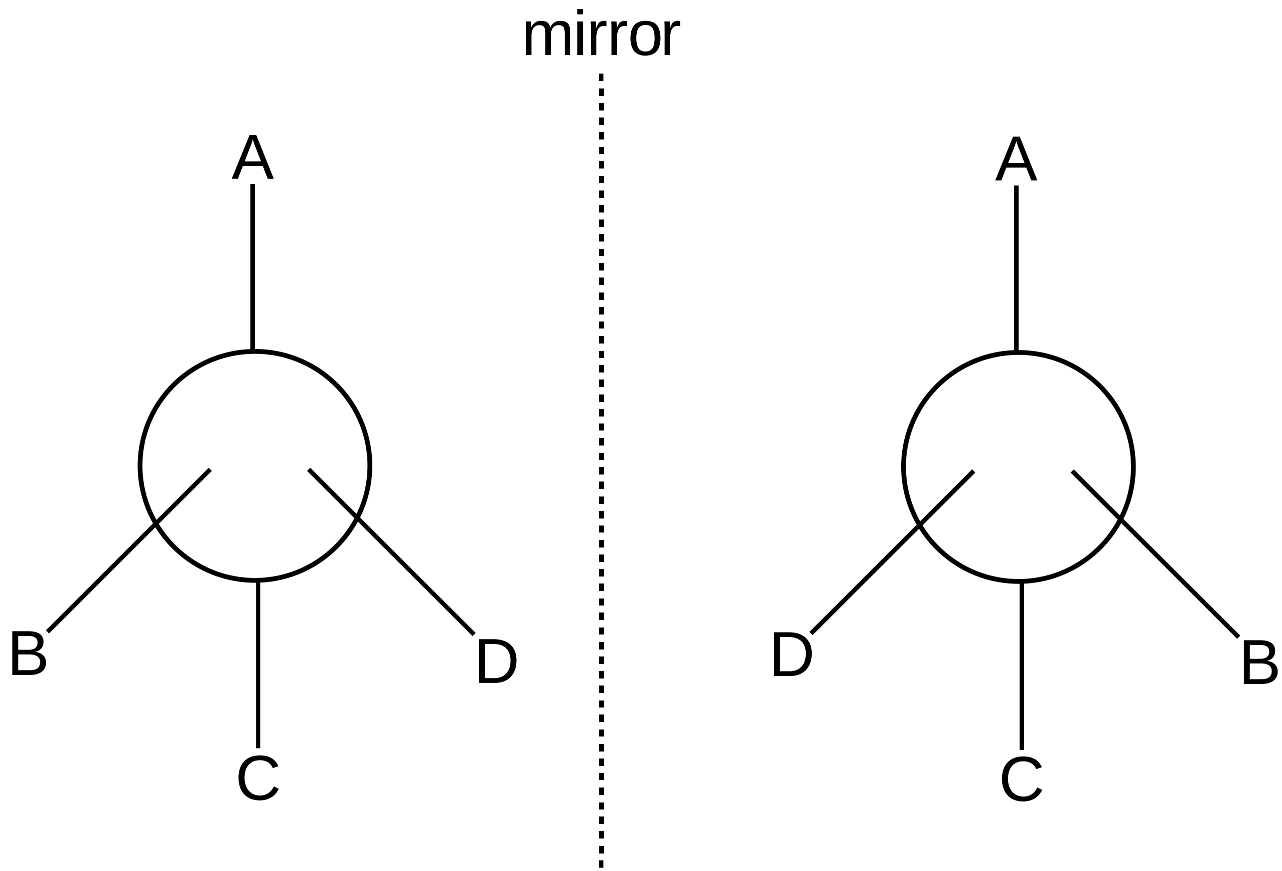
Some objects are not the same as their mirror images (technically, they have no plane of symmetry)

A right-hand glove is different than a left-hand glove. The property is commonly called “handedness”



- A molecule (or object) that is *not* superimposable on its mirror image is said to be *chiral.*

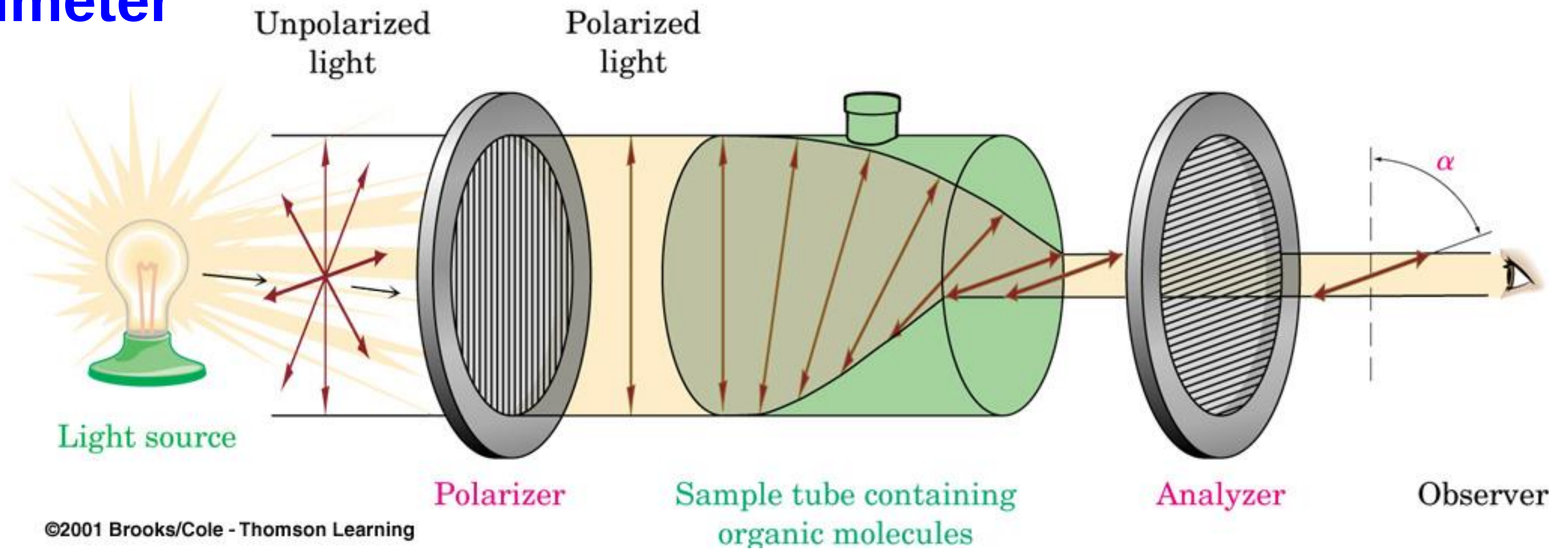
The two are non-superimposable, mirror images. Such isomers are called enantiomers.



Optical Activity - molecules enriched in an enantiomer will rotate plane polarized light are said to be *optically active*. The optical rotation is dependent upon the substance, the concentration, the path length through the sample, and the wavelength of light.

Polarimeter

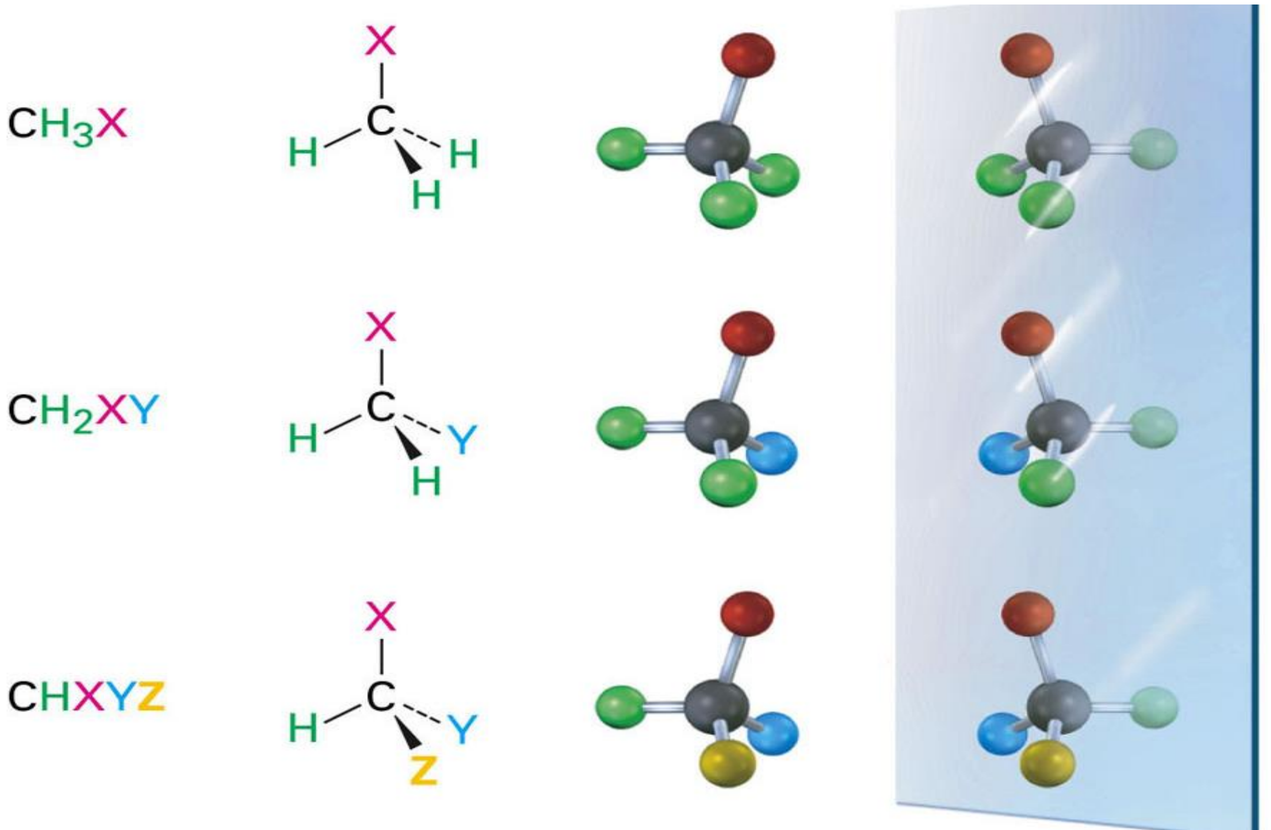
589 nm - D-line of a sodium lamp



Plane polarized light: light that oscillates in only one plane

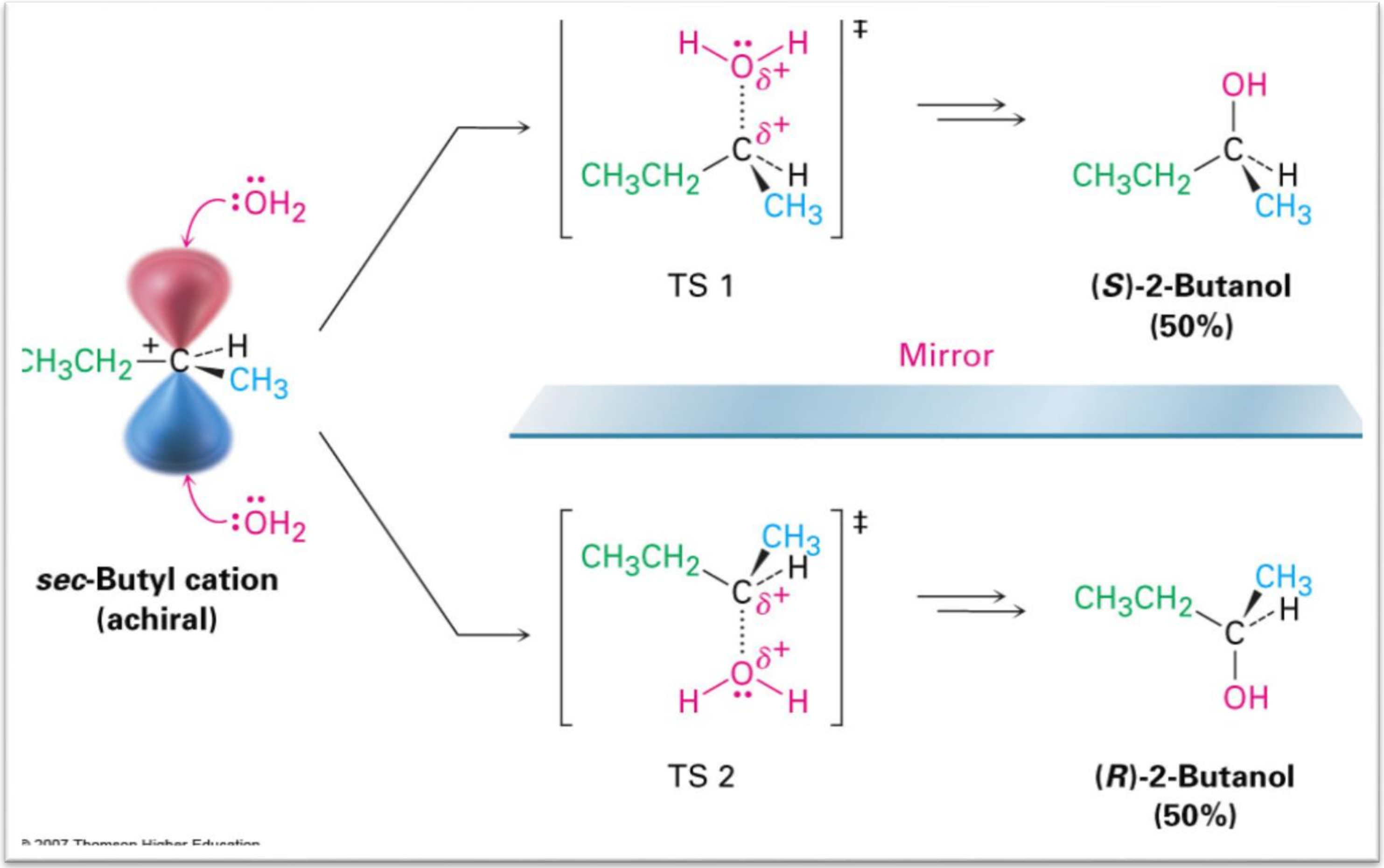
Examples of Enantiomers

- Molecules that have one carbon with 4 different substituents have a non-superimposable mirror image
- **Enantiomers** = non-superimposable mirror image stereoisomers
- Enantiomers have identical physical properties (except for one)
- Build molecular models to see this



Achiral Intermediate Gives Racemic Product

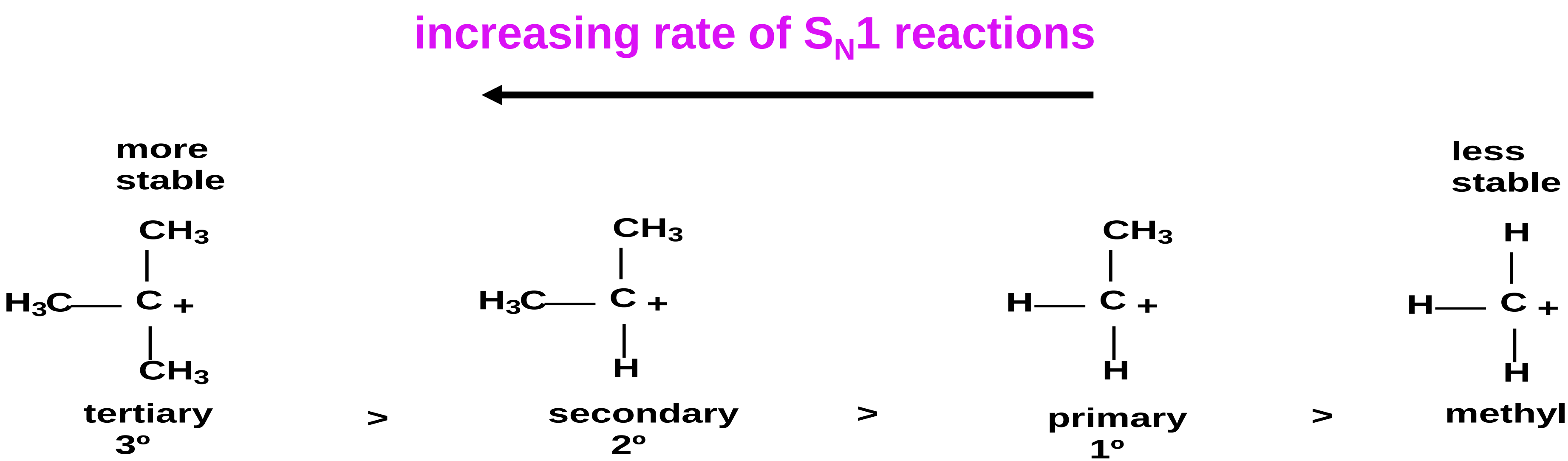
- Addition via carbocation
- Top and bottom are equally accessible
- Achiral reactant + Achiral reactant = Optically Inactive Product
- Optical Activity doesn't come from nowhere

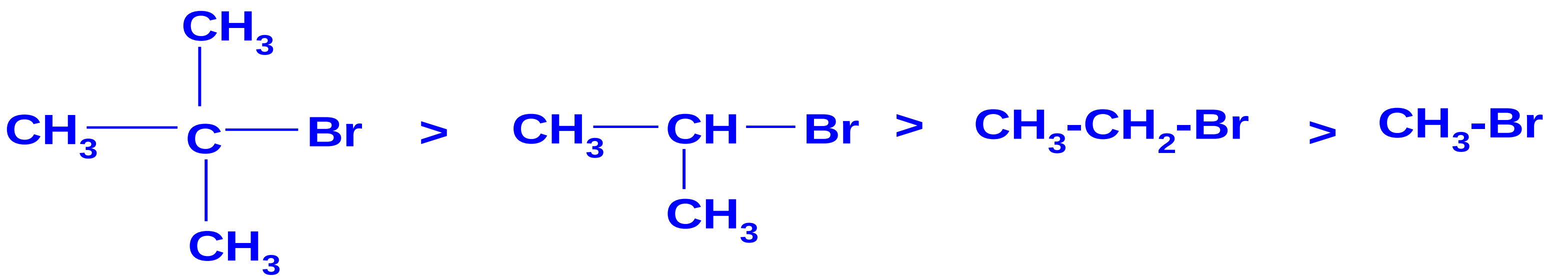


- The unimolecular reaction proceeds through an intermediate carbocation which is planer (sp2 hybridized C atom) . It is therefore expected that a chiral alkyl halide will give a racemic alcohol since the carbocation has an equal chance of being attacked by the nucleophillic reagents from either side of the plane.
- The total racemization takes place in SN1 reaction. If the attack of the nucleophile on the two faces are purely random , it would expect to obtain equal amount of two enantiomers (racemic mixture)
- An optically pure (+) or (-) substrate give a product which is only about 50% or less optically pure. In other words, the starting material contains only one enantiomer where as the product contain both.

Factors influencing Rate of S_N1 Reaction

- Rate of the reaction does not depend on the concentration of the nucleophile.
- It depends only on the concentration of the substrate.
- Reaction is fastest when the alkyl group of the substrate is tertiary and intermediate for secondary and slower for primary.
- Because formed carbocation is more stable in tertiary carbon compared to primary. $3° > 2° > 1°$

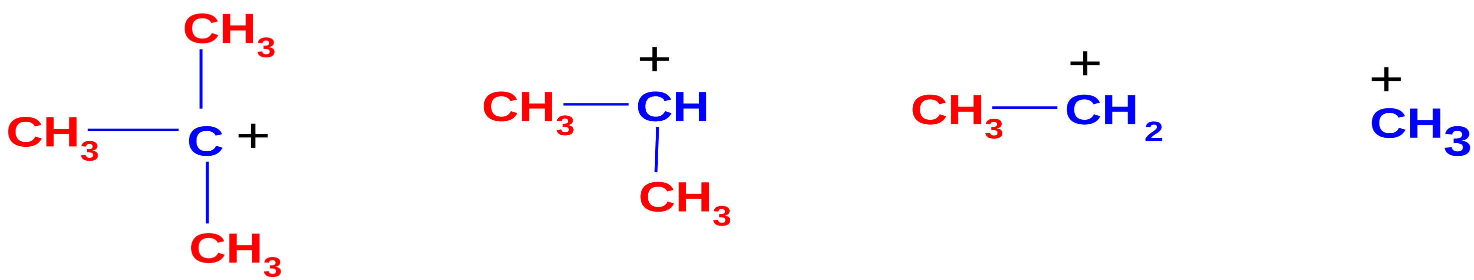




tertiary

secondary

primary



tertiary carbocation (very stable)

three methyl groups

secondary carbocation

two methyl groups

primary carbocation (unstable)

one methyl group

very unstable carbocation

no methyl groups

Effects of Leaving Group

- An SN1 reaction speeds up with a good leaving group (leaving group is involved in the rate-determining step)

- A good leaving group wants to leave so it breaks the C-Leaving Group bond faster. Once the bond breaks, the carbocation is formed and the faster the carbocation is formed, the faster the nucleophile can come in and the faster the reaction will be completed.

- A good leaving group is a weak base because weak bases can hold the charge. They're happy to leave with both electrons and in order for the leaving group to leave, it needs to be able to accept electrons.

- Strong bases, on the other hand, donate electrons which is why they can't be good leaving groups.

As you go from left to right on the periodic table, electron donating ability decreases and thus ability to be a good leaving group increases.

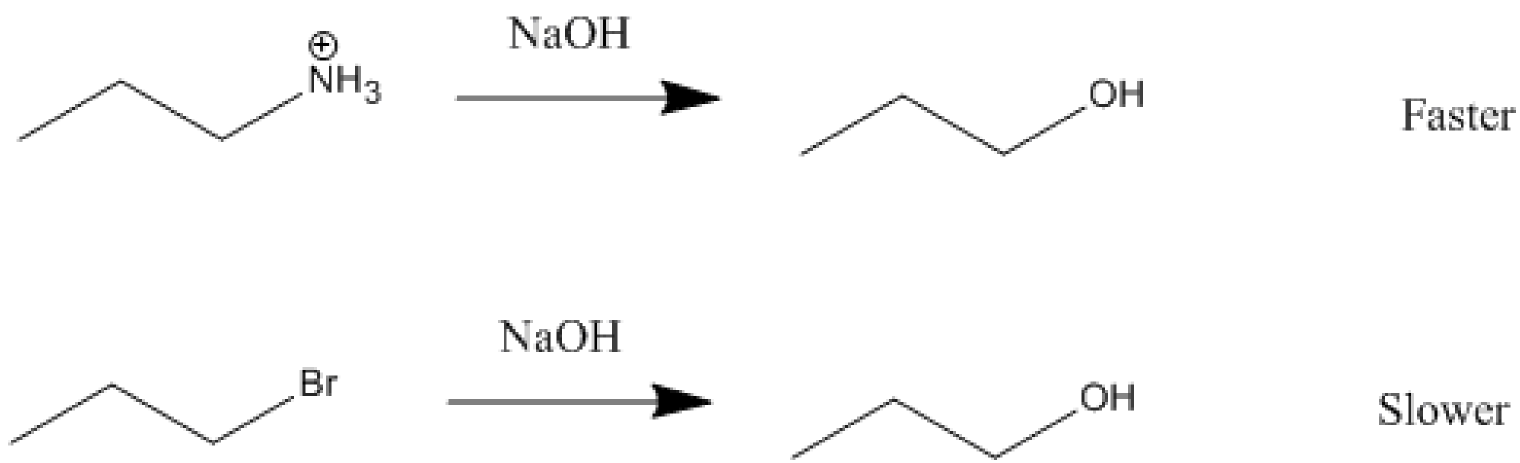
Halides are an example of a good leaving group whos leaving-group ability increases as you go down the column

$I^- > Br^- > Cl^- > F^-$

Best Worst

(para-toluenesulfonate = "tosylate", TsO−).

The two reactions below is the same reaction done with two different leaving groups. One is significantly faster than the other. This is because the better leaving group leaves faster and thus the reaction can proceed faster.



Effect of Solvent

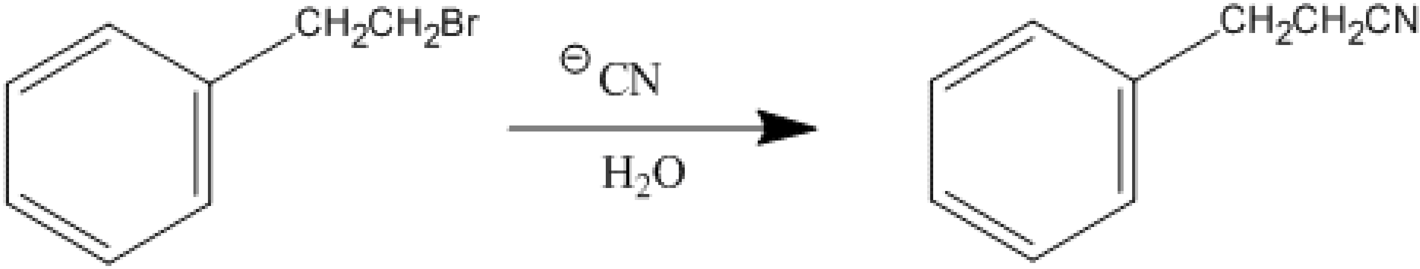
- Sometimes in SN1 reactions, the solvent acts as the nucleophile (solvolysis reaction)

$$RX + H_2O \xrightarrow[\text{water}]{} ROH + HX$$

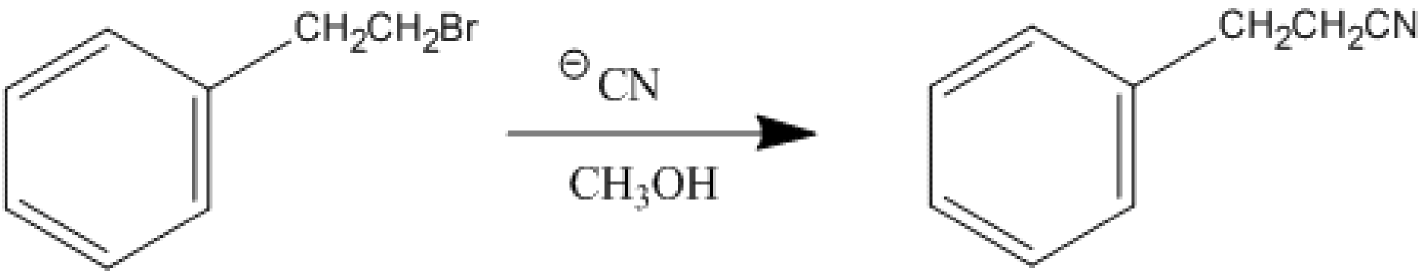
- The polarity and the ability of the solvent to stabilize the intermediate carbocation is very important
- A dielectric constant below 15 is usually considered non-polar. Basically, the dielectric constant can be thought of as the solvent's ability to reduce the internal charge of the solvent.
- Higher the dielectric constant the more polar the substance and in the case of SN1 reactions, the faster the rate.

Solvent	Dielectric Constant	Relative Rate
CH_3CO_2H	6	1
CH_3OH	33	4
H_2O	78	150,000

Below is the same reaction conducted in two different solvents and the relative rate that corresponds with it.



Relative Rate: 150,000



Relative Rate: 4

polar protic solvents are: acetic acid, isopropanol, ethanol, methanol, formic acid, water, etc

Solvolysis

A nucleophilic substitution reaction where the solvent is the nucleophile is called solvolysis.

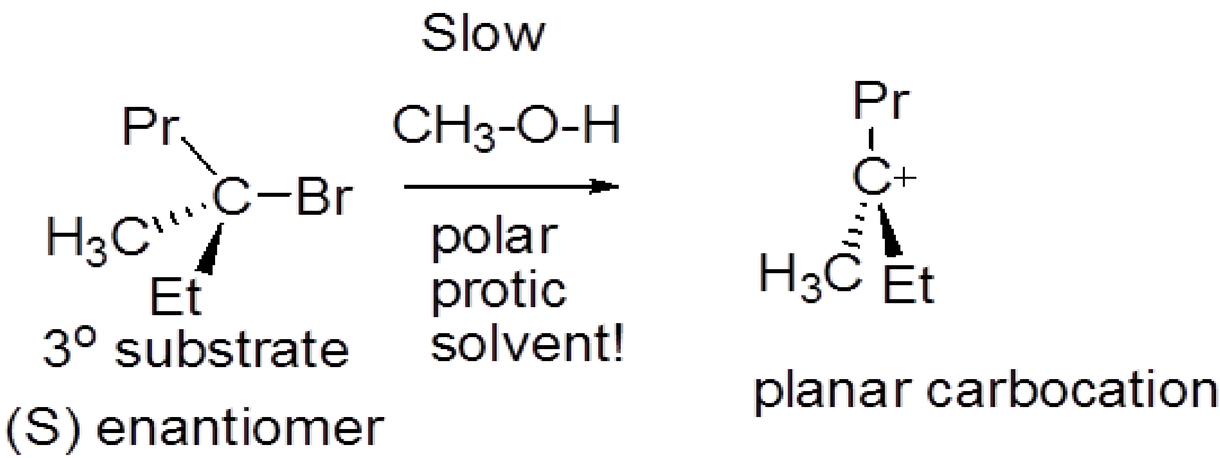
When water is the solvent, the term hydrolysis is used.

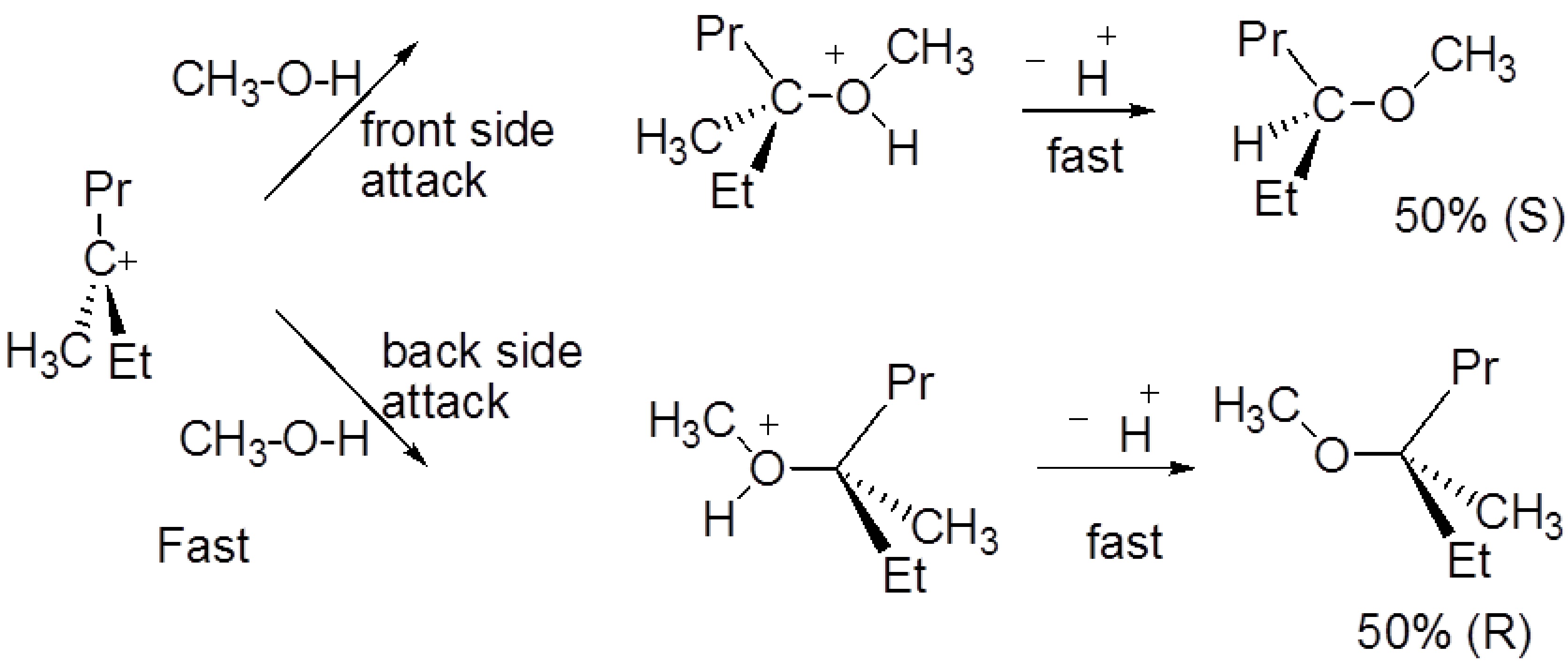
Methanolysis means methanol is the nucleophile/solvent.

$$RX + H_2O \xrightarrow[\text{water}]{} ROH + HX \qquad \textit{hydrolysis}$$

$$RX + CH_3OH \xrightarrow[\text{methanol}]{} ROCH_3 + HX \qquad \textit{methanolysis}$$

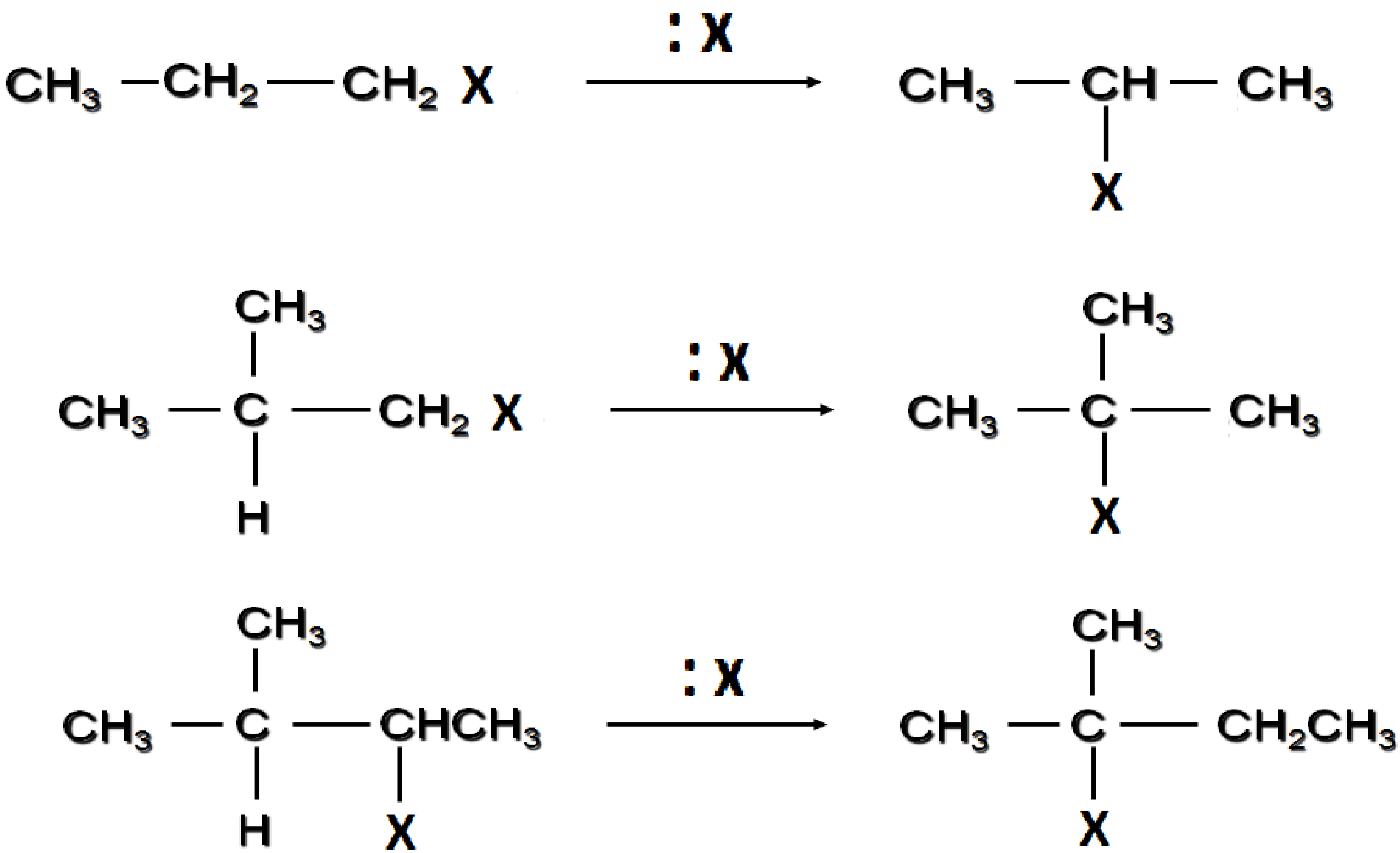
$$RX + CH_3CH_2OH \xrightarrow[\text{ethanol}]{} ROCH_2CH_3 + HX \qquad \textit{ethanolysis}$$


Slow
CH3-O-H
polar
protic
solvent!
Pr
C
Br
H3C
Et
3° substrate
(S) enantiomer
C+
planar carbocation

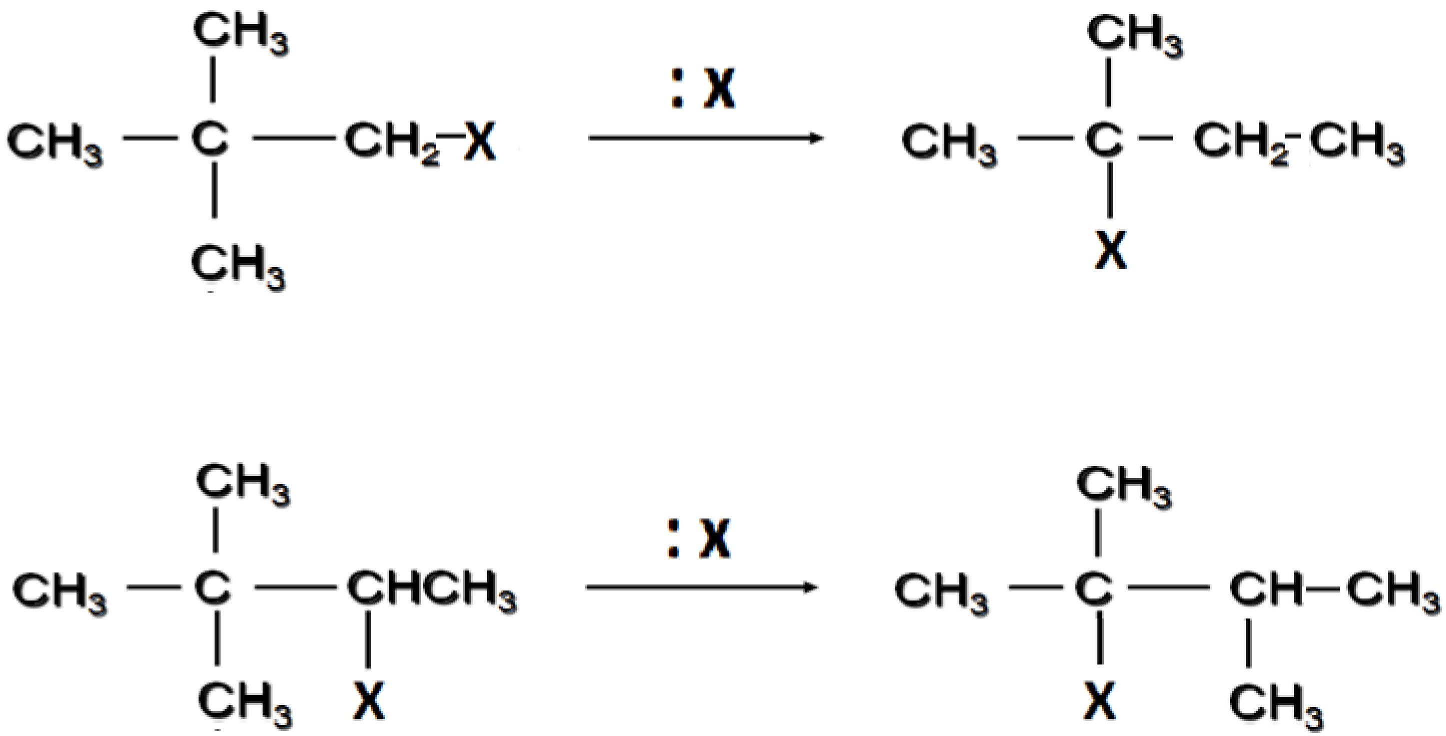

CH3-O-H
front side
attack
back side
attack
Fast
Pr
C+
H3C
Et
CH3
O
H
− H+
fast
50% (S)
50% (R)

Carbocation Rearrangements in S_N1 Reactions

- It is observed that the entering group attached to a different carbon atom than one that of original position.

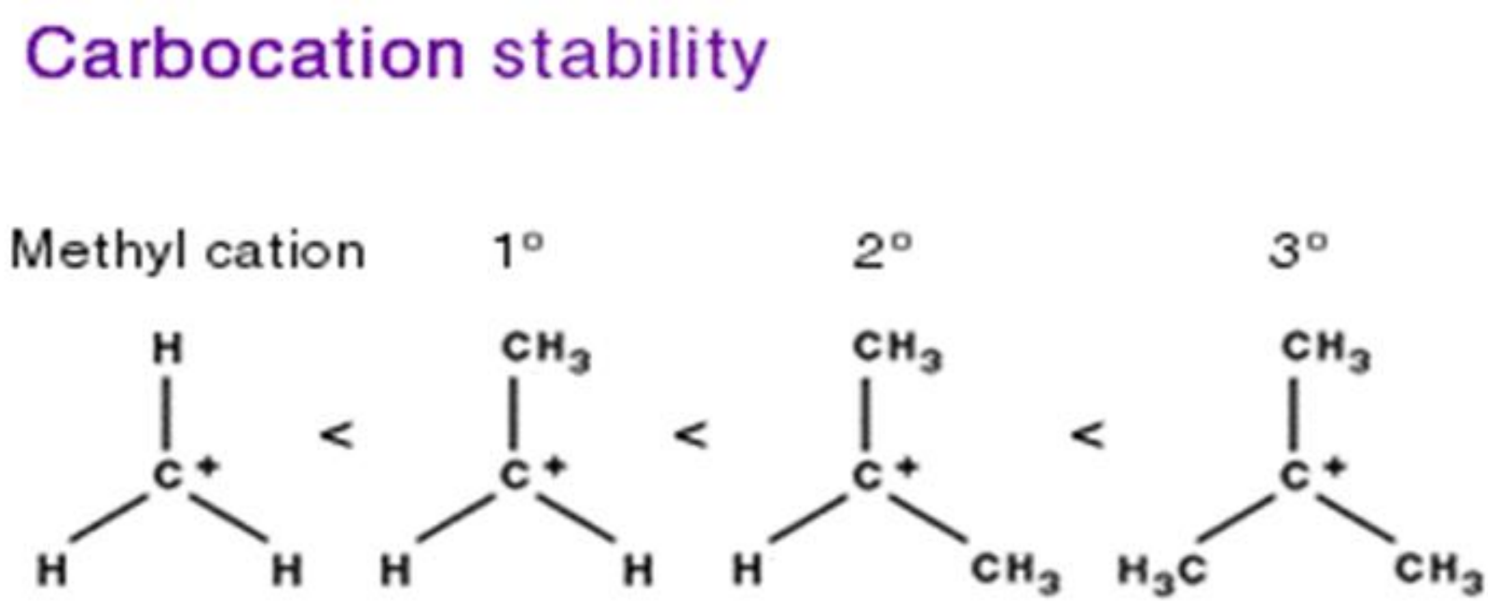


Rearrangement in the C skeleton

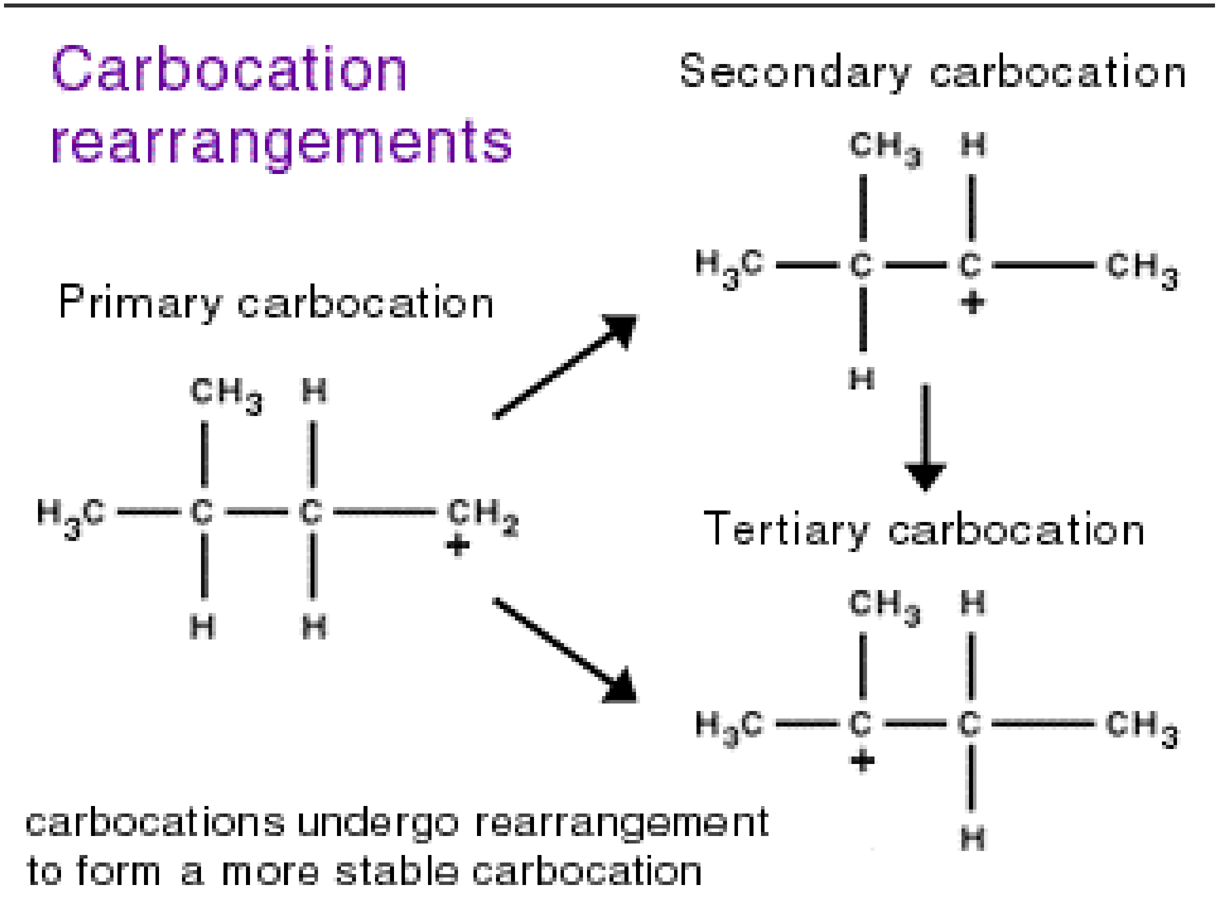


- Carbocations are intermediates in S_N1 reactions, therefore rearrangements are possible
- Less stable carbocations converted to more stable forms by migrating ‘H’ atoms, alkyl group with an electron pair from an adjacent ‘C’ atom, to the C having the +ve charge..

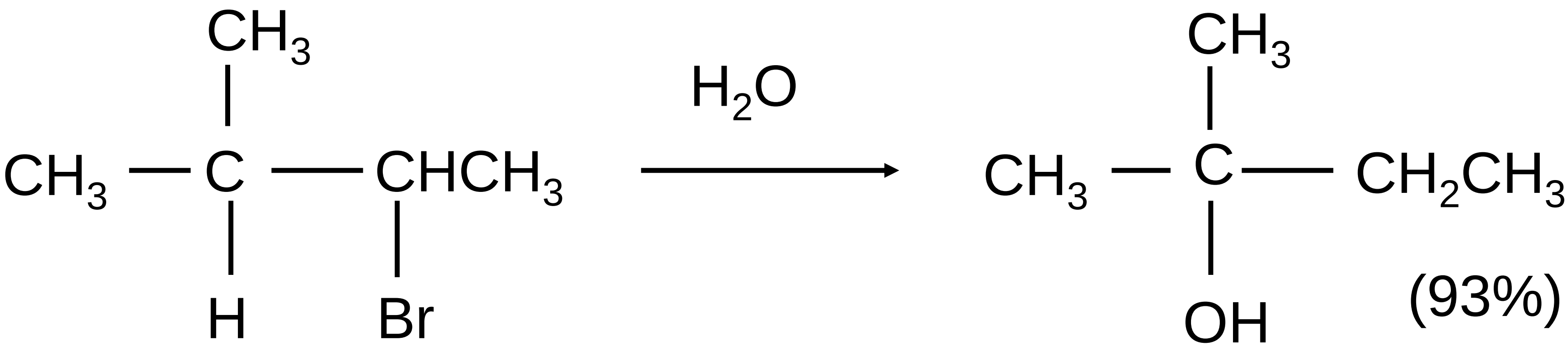
Migration of ‘H’ atom = hydride shift
Migration of ‘C’ atom = alkyl shift



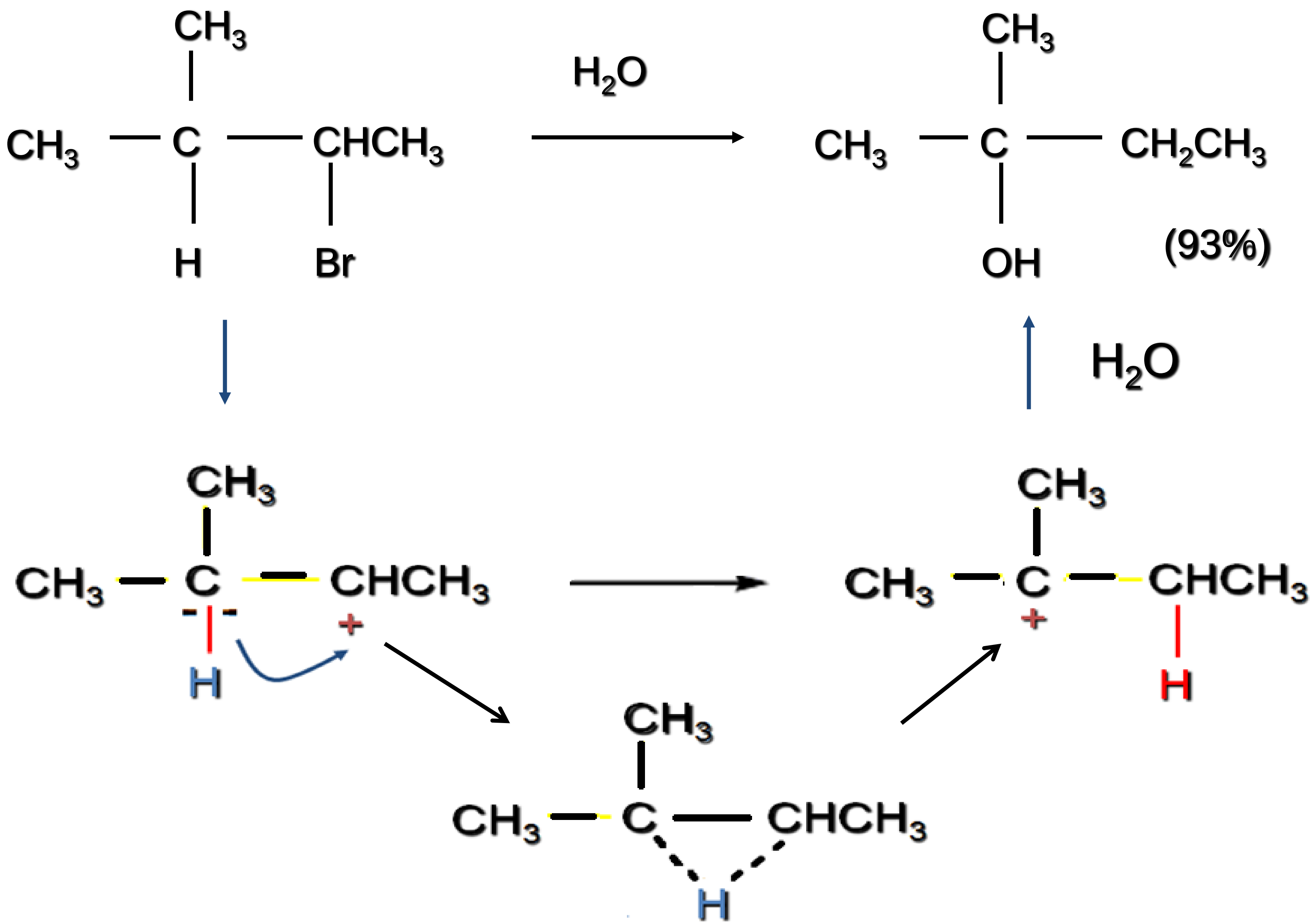
Carbocations rearrange to the more stable form(s)

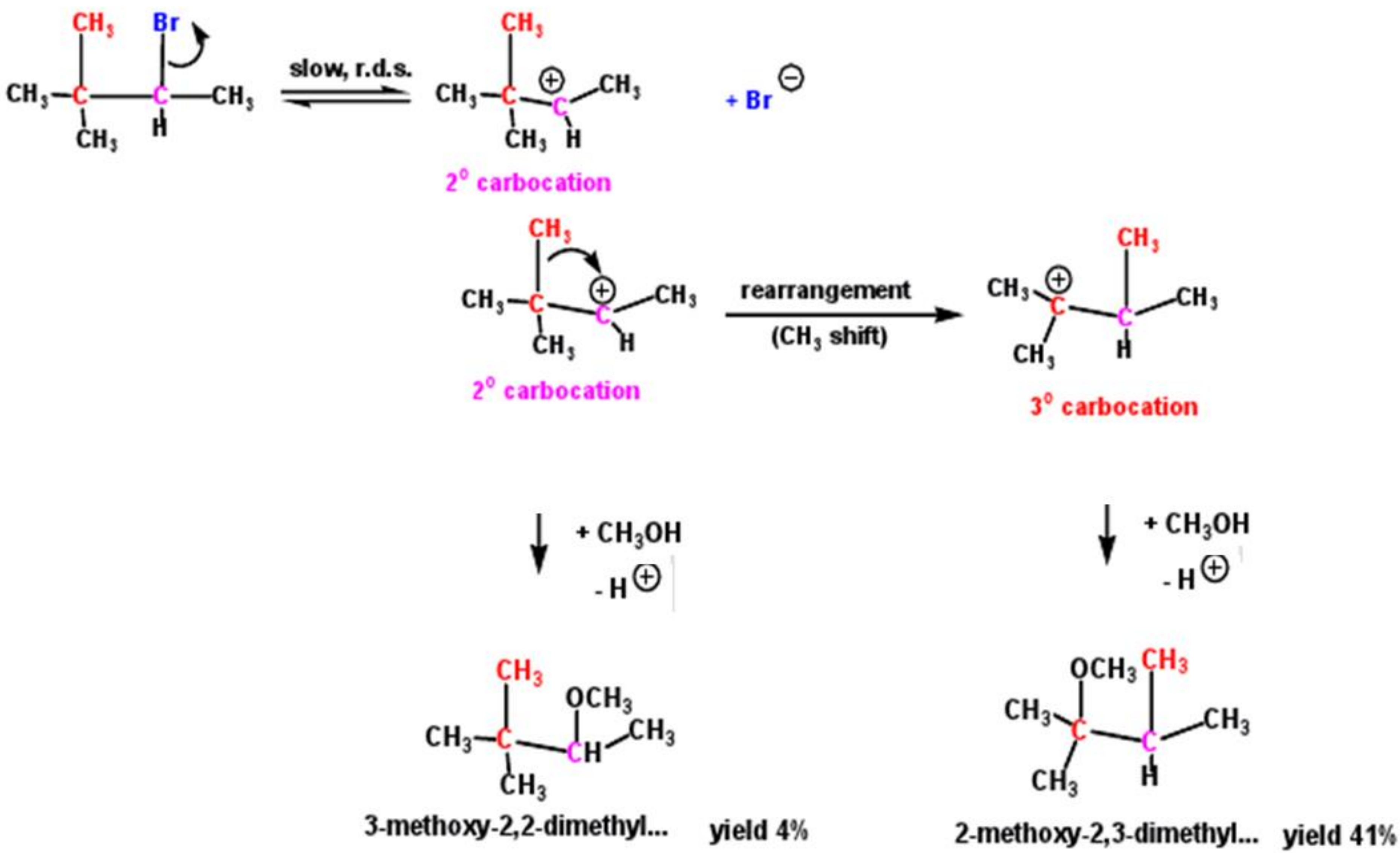


Example



Example


CH3
CH3 — C — CHCH3
H Br
H2O
CH3
CH3 — C — CH2CH3
OH
(93%)
H2O
CH3
CH3 — C — CHCH3
H
+
CH3
CH3 — C — CHCH3
H
CH3
CH3 — C — CHCH3
+
H

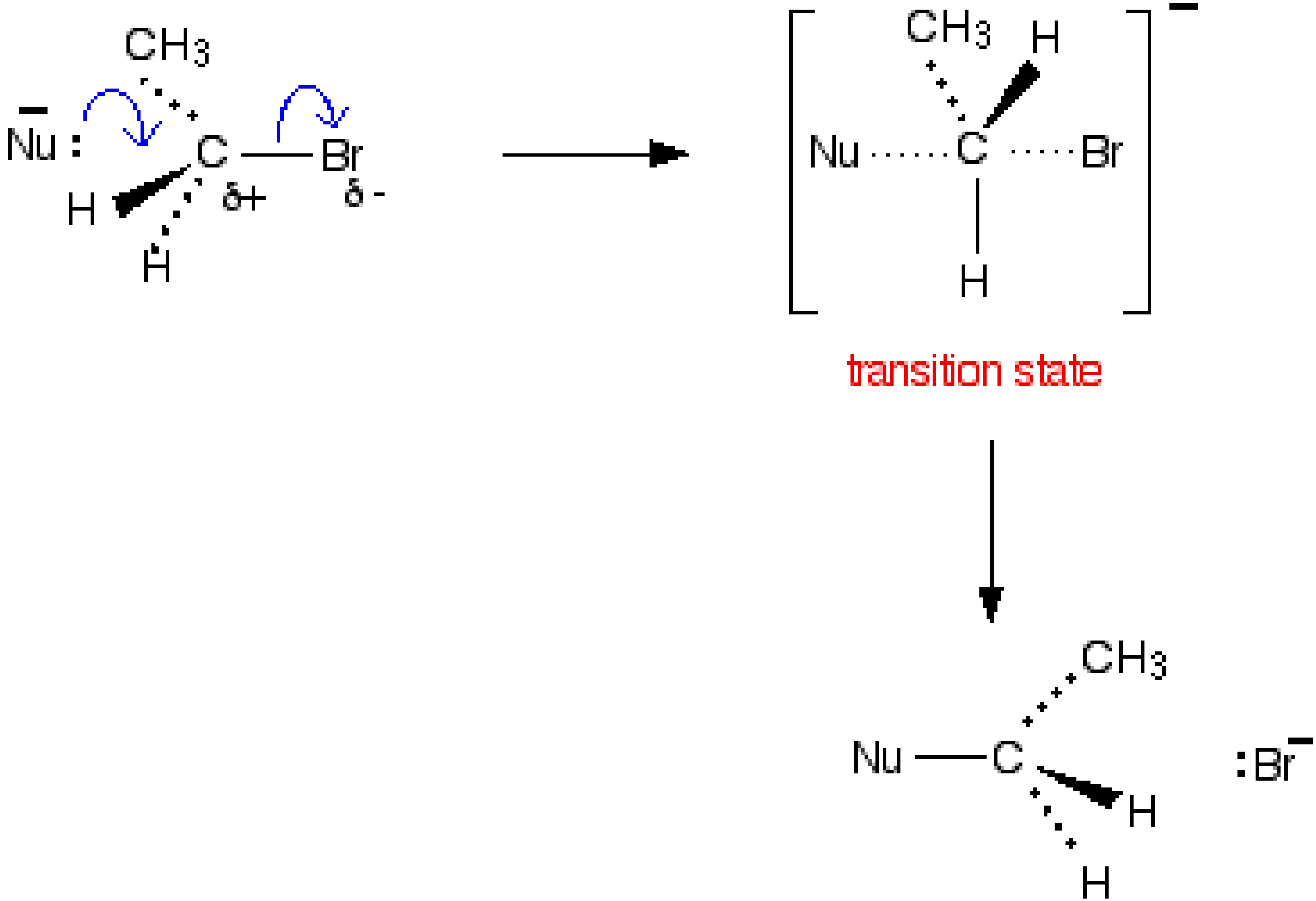


slow, r.d.s.
+ Br
2° carbocation
2° carbocation
rearrangement
(CH3 shift)
3° carbocation
+ CH3OH
- H
+ CH3OH
- H
3-methoxy-2,2-dimethyl...
yield 4%
2-methoxy-2,3-dimethyl...
yield 41%

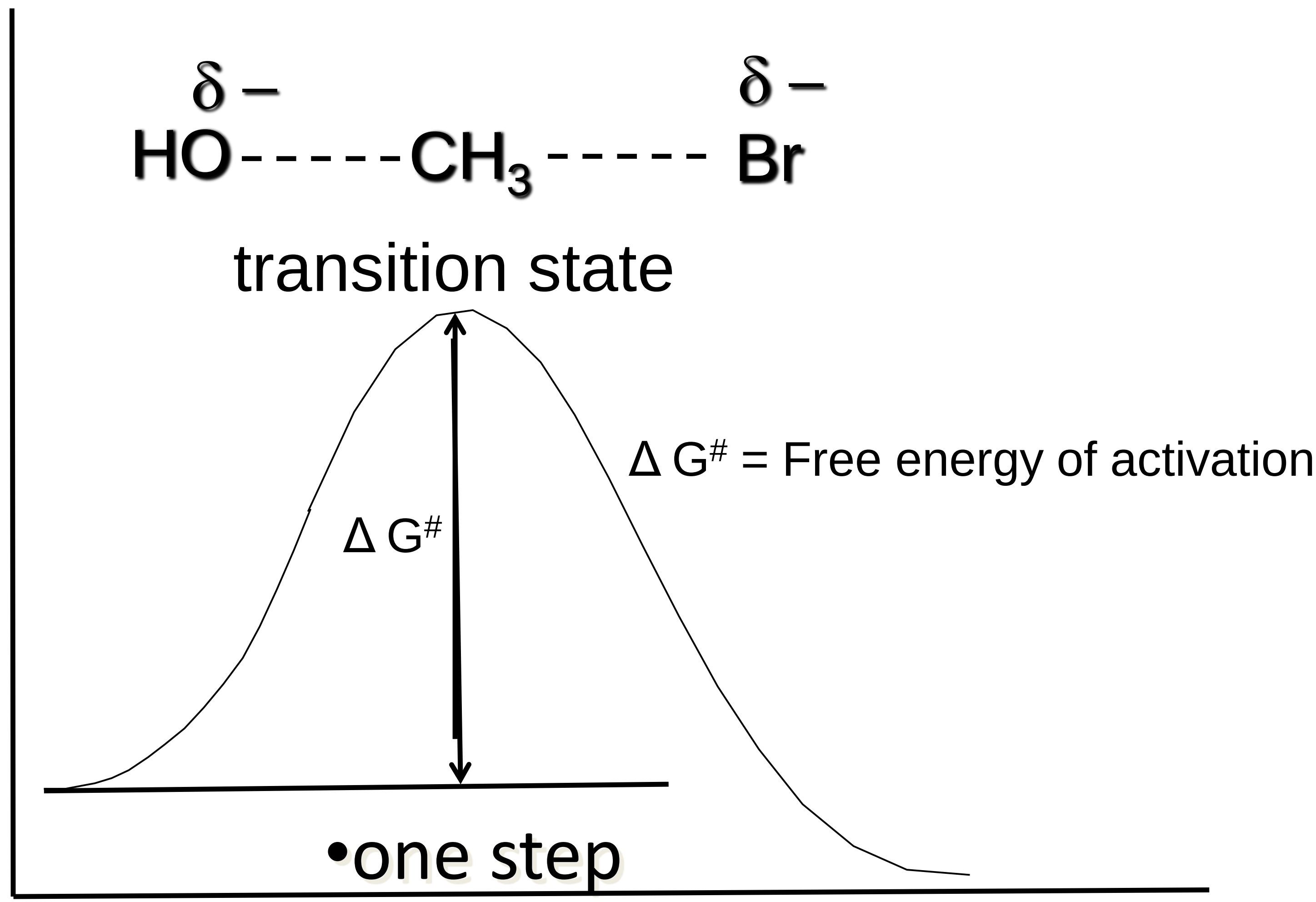
Nucleophilic Substitution S_N2 Mechanism

- "S & N" stands for 2 indicates the reaction is bimolecular i.e. the nucleophile and the substrate are involved in the key step.
- It's a one step process.
- The nucleophile attacks from the back side of C – L bond.
- Nucleophile and the leaving group partly bonded to the carbon at transition state (no intermediates).
- Then departure of the leaving group.

S_N2 Mechanism



Bimolecular mechanism



$$HO^{-} + CH_3Br \longrightarrow HOCH_3 + Br^{-}$$

$$HO^{-} + \underbrace{CH_3Cl}_{\text{Methyl Chloride}} \longrightarrow CH_3\text{-}OH + Cl^{-}$$

$$\downarrow \qquad\qquad\qquad\qquad \uparrow$$

$$HO^{-} + CH_3\text{—}Cl \longrightarrow [\,HO\text{----}CH_3\text{----}Cl\,]^{\pm}$$

$$\text{Rate} = K_r \quad = \quad [\text{MeCl}]\,[\text{OH}]$$

The reaction rate depend on the concentration of both molecules.

.

Factors influencing Rate of S_N2 Reaction

- Concentration of the nucleophile and concentration of substrate increases the rate

Factors influencing Rate of S_N2 Reaction….

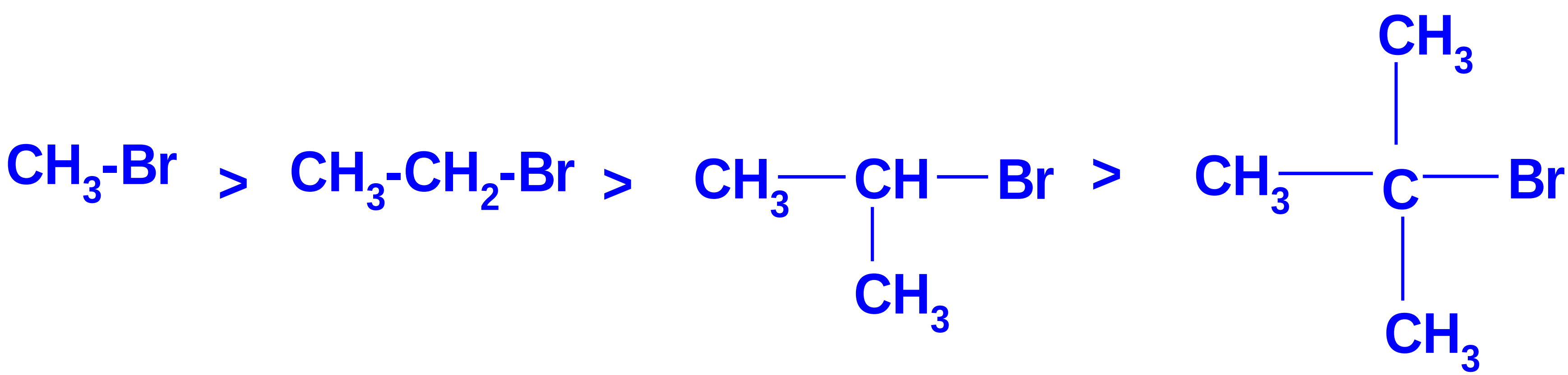
- The bimolecular mechanism proceeds through a highly sterically hindered transition state, where five groups are bonded to the carbon. Therefore the larger groups causes higher strain energy, slower the reaction. Reaction is fastest when the alkyl halide is primary; intermediate for secondary and slower for tertiary.

  Because the formation of the transition state, it is easy for primary compare to tertiary.

  $1^\circ > 2^\circ > 3^\circ$

Reactivity toward substitution by the SN2 mechanism

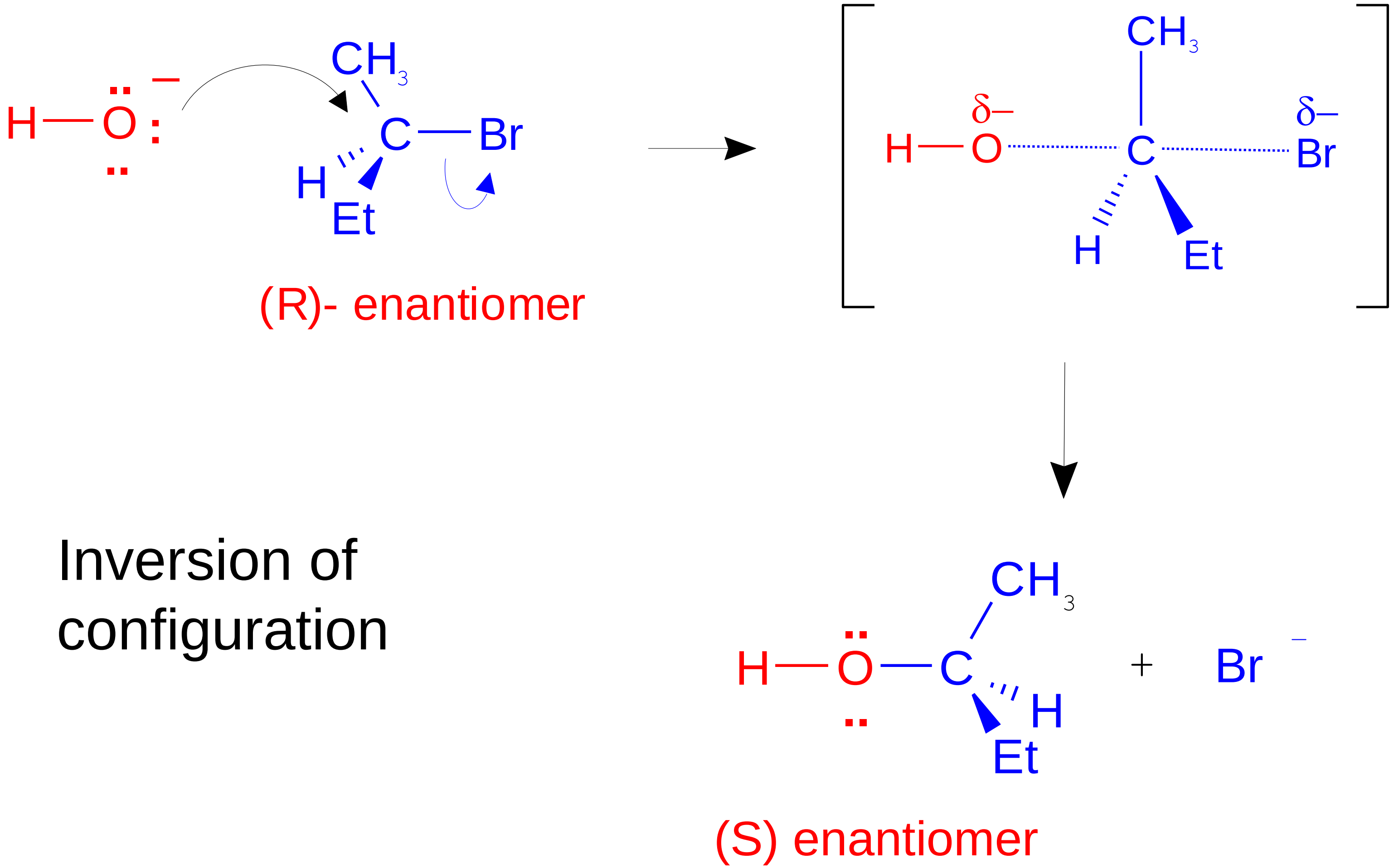
Alkyl bromide	Class	Relative rate
• CH_3Br	Methyl	221,000
• CH_3CH_2Br	Primary	1,350
• $(CH_3)_2CHBr$	Secondary	1
• $(CH_3)_3CBr$	Tertiary	too small to measure



primary secondary tertiary

S_N2 Reaction: stereochemistry

Nucleophilic substitutions $S_N2$are stereospecific and proceed with inversion of configuration.

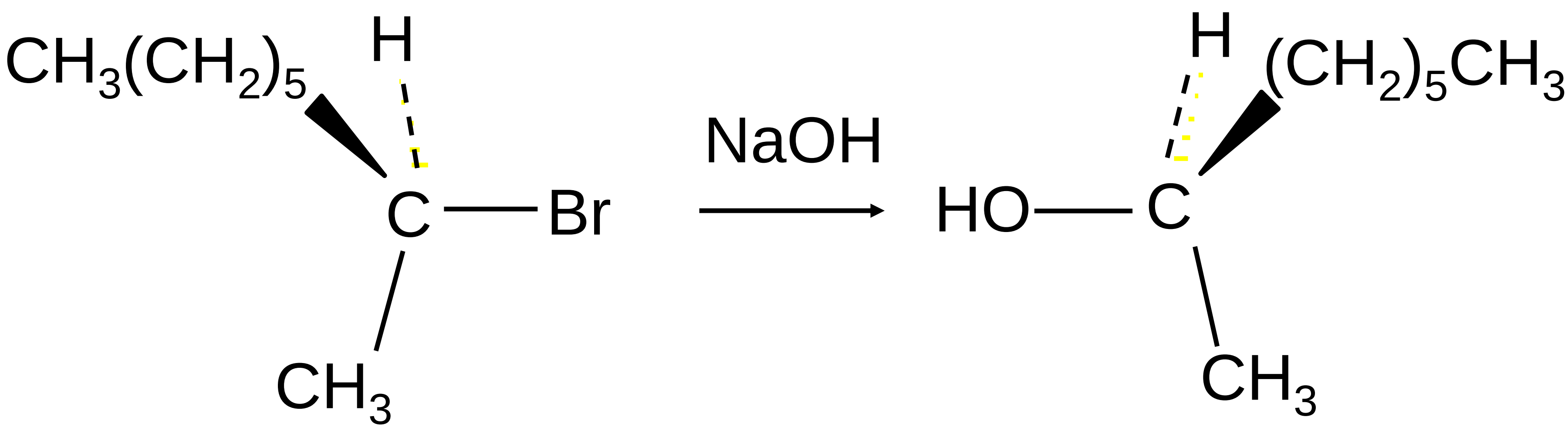


Inversion of configuration

- The reaction of 2-bromooctane with NaOH (in ethanol-water) is stereospecific.

(+)-2-Bromooctane → (–)-2-Octanol

(–)-2-Bromooctane → (+)-2-Octanol



(*S*)-(+)-2-Bromooctane (*R*)-(–)-2-Octanol

S_N1 and S_N2 Comparison

- Polar protic solvents favor S_N1 reaction

  e.g. H_2O
- Polar aprotic solvent favor S_N2 reaction

  e.g. Acetone, DMF, DMSO
- If the nucleophile is strong; then S_N2 is favored.

  1. HO- > HOH
  2. HS- > HO-
  3. $NH_3 > H_2O > HF$

S_N1 and S_N2 Comparison

- If the carbon bearing the leaving group is stereogenic:
  1. S_N1 reaction occurs with loss of optical activity (that is, with racemization)
  2. S_N2 reaction occurs with inversion of configuration.